

# MTH-210: End-Semester Examination

Monsoon-2025

December 8, 2025

**Full Marks: 70**

**Time: 3:00 hours**

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**General Instructions:** This is a closed book examination. Please be precise, brief and to the point in your answer. Unnecessary blabbers or unreadable handwriting will fetch deduction of marks. Also, if you want to disprove a statement by a counterexample, then your counterexample must be explicit. In addition, your counterexample must be complete and provide a clear explanation why it is a counterexample. Vague ideas and reasonings will not incur full credits.

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1. Let  $B$  be a set and  $S_1$  and  $S_2$  be two distinct equivalence relations on  $B$ . Define  $S_1 \oplus S_2 = (S_1 \cup S_2) - (S_1 \cap S_2)$ . Then, prove or disprove the following statements.
  - (a)  $S_1 \oplus S_2$  is an equivalence relation.
  - (b)  $S_1 \oplus S_2$  is a symmetric relation. (5 Marks + 5 Marks)
2. Let  $S_1$  be a partial order on  $A$  and  $S_2$  be a partial order on  $B$ . Define a relation  $S$  on  $A \times B$  as follows:  $((a, b), (c, d)) \in S$  if and only if  $(a, c) \in S_1$  and  $(b, d) \in S_2$ .  
Prove or disprove: if  $S_1$  is a total order on  $A$  and  $S_2$  is a total order on  $B$ , then  $S$  is a total order on  $A \times B$ . (8 Marks)
3. Let  $B$  be the set of all real numbers that do not have 2 in their decimal representation. Then, prove or disprove that  $B$  is countable. (8 Marks)
4. A relation  $S$  on a set  $B$  is *circular* if  $(a, b) \in S$  and  $(b, c) \in S$ , then  $(c, a) \in S$ . Prove or disprove: if  $S$  is circular and reflexive, then  $S$  is an equivalence relation. (8 Marks)
5. In the  $xy$ -plane, consider two coordinates  $(0, 0)$  and  $(m, n)$  for some integers  $m, n > 0$ . A path from  $(0, 0)$  to  $(m, n)$  is a *lattice path* if the path is made up of a series of steps, where each step is a move one unit to the right or a move one unit upward. No move to the left or downward is allowed. Explain by using a counting argument how many lattice paths are there from  $(0, 0)$  to  $(m, n)$ . (6 Marks)
6. Let  $n$  and  $k$  be positive integers such that  $n \geq k$ . Prove by using combinatorial argument that  $\sum_{k=1}^n k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$ . (8 Marks)

[N.B. If you prove by algebraic manipulation, then you will be awarded zero marks. Your proof must be double-counting argument or bijective proof]

7. If  $n$  and  $d$  are positive integers, then  $n(\bmod d)$  denotes the remainder when  $n$  is divided by  $d$ . Prove or disprove that: if a set of 20 ordered pairs are chosen from  $\mathbb{Z}^+ \times \mathbb{Z}^+$ , then there are 3 distinct sets  $\{(a_1, a_2), (b_1, b_2)\}$ ,  $\{(p_1, p_2), (q_1, q_2)\}$ , and  $\{(d_1, d_2), (r_1, r_2)\}$  such that

- $a_1(\bmod 3) = b_1(\bmod 3)$ ,  $a_2(\bmod 3) = b_2(\bmod 3)$ ,
- $p_1(\bmod 3) = q_1(\bmod 3)$ ,  $p_2(\bmod 3) = q_2(\bmod 3)$ , and
- $d_1(\bmod 3) = r_1(\bmod 3)$ ,  $d_2(\bmod 3) = r_2(\bmod 3)$ .

**(8 Marks)**

[N.B.  $37(\bmod 5) = 2$  because if 37 is divided by 5, then we get that the quotient is 7 and the remainder is 2.]

8. Let  $G = (V, E)$  be a simple graph with  $n$  vertices and  $G$  has an edge whose removal makes it a tree. Let  $V_1$  denote the set of all pendant vertices,  $V_2$  denote the set of all vertices with degree 2, and  $V_3$  denote the vertices of  $G$  with degree at least 3. Then, prove or disprove that  $|V_3| \leq |V_1|$ . **(7 Marks)**

9. Let  $f : A \rightarrow B$  be a well-defined function. For a subset  $P \subseteq A$ ,  $f(P) = \{y \in B \mid y = f(x) \text{ for some } x \in P\}$ . Then, prove or disprove: for every  $D_1, D_2 \subseteq A$ ,  $f(D_1) \cap f(D_2) = f(D_1 \cap D_2)$ . **(7 Marks)**

## **APPENDIX:**

- The set  $\mathbb{Z}^+$  denotes the set of all positive integers which is the same set as  $\mathbb{N} - \{0\}$ . Equivalently,  $\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$ .
- If you want to prove that an infinite set is countable, then please prove the existence of a precise bijection. If you want to prove that an infinite set is uncountable, then you can use diagonalization and some properties of uncountable sets.
- An undirected graph is a *simple graph* if it has no self loop or parallel edges.
- An undirected graph is *connected* if there is a simple path between every pair of vertices.
- A *cycle* is a simple path that starts and ends at the same vertex.
- A *tree* is an undirected graph that is connected but has no cycle.
- Let  $G = (V, E)$  be a graph and  $B \subseteq E$ . Then,  $G - B$  denotes the graph that is obtained by deleting the edges that are in  $B$ . Formally,  $G - B = (V, E - B)$  which is the graph with vertex set  $V$  (same as the vertex set of  $G$ ) and the edge set  $E - B$ .
- You can use any other lemma/theorem that was proved in the class.